

ELT3207 Meas. & Digital Control Basics

Lecture 6: Random error analysis

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Introduction

- ❑ Up to now, we have finished the implementation of digital measurement systems. Is the job done?
 - **Random features**: even if the same measuring system is used to measure a fixed measurand repeatedly, the results will not have all the same value.
 - In some cases, the randomness in the data is so dominant that it is **difficult** to define what is the value are we reading from the set of confusing data.
- ❑ Statistics offer tools to **interpret** such sets of confused data
 - To identify **trends** that are useful for **planning** measurements
 - To identify & generalize the characteristics of measurement data
 - E.g., is there a significant chance that the temperature will ever exceed 1117°C?

TABLE 6.1 Results of 60 Temperature Measurements in a Duct

Number of readings	Temperature (°C)
1	1089
1	1092
2	1094
4	1095
8	1098
9	1100
12	1104
4	1105
5	1107
5	1108
4	1110
3	1112
2	1115

Statistics revisit : general concepts (1)

❑ Population

- The **entire** collection of objects, measurements, observations, etc.

❑ Sample

- A **representative subset** of a population.
- Although it is nice if we know the whole population, often in reality what we can observe is only a sample due to constraints in time, cost, ...

❑ Sample space

- The set of all possible samples can be drawn from a population.

❑ Random variable

- A **numerically valued function** defined for the population, i.e. for every possible sample point, there is a corresponding numerical value.
- A random variable can be continuous or discrete

❑ Distribution function

- The mathematical relationship that is used to represent the values of the random variable.

Statistics revisit: general concepts (2)

❑ Parameter

- A numerical attribute of the entire **population**, example: the average value of a random variable in a population is a parameter for that population

❑ Event

- The outcome of a random experiment.

❑ Statistic

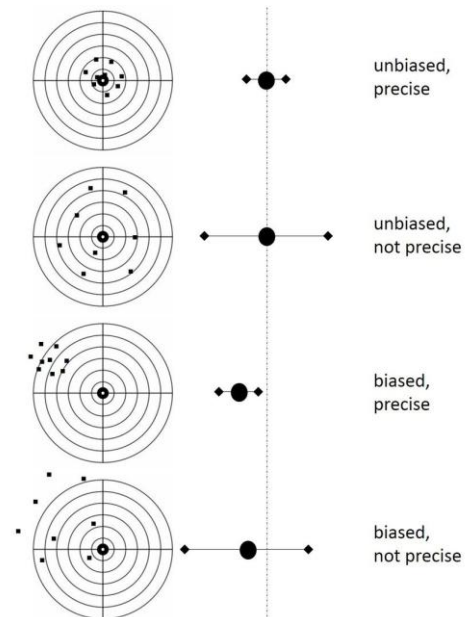
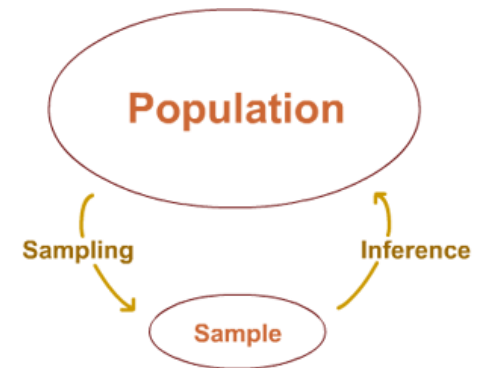
- A numerical attribute of the **sample**, e.g. the average value of a sample property is a statistic of the sample

❑ Confidence interval & confident level

- **Confidence interval**: a range of estimates for an unknown parameter: $\bar{x} \pm u_{\bar{x}}$.
- **Confidence level**: the probability (or long-run frequency) of confidence intervals that contain the true value of the parameter: $x' = \bar{x} \pm u_{\bar{x}} (P\%)$.
- **Level of significance (α)**: the probability that the mean will fall outside the confidence interval, i.e. confidence level = $1 - \alpha$.

Statistical analysis methodology

- ❑ Objective: use information from a sample to **predict** the behavior of a population → statistical **inferences**.
 - Statistical methods can help you make the “best educated guess” about a property of the population.
- ❑ Sample data are used to construct **estimates** for the **parameter** of the population that we are interested in.
 - Estimates vary from sample to sample → a good estimator must be **unbiased** and precise, i.e. the **central tendency** of the estimates must be the same with the parameter of interest while the **dispersion** from the central tendency is low.
 - In most useful sampling schemes, variability from sample to sample can be estimated via a single sample.



Fundamental statistics

- ❑ Steps to apply statistical analysis to measurement data:
 - Determine **statistics** of the sample (e.g. central tendency, dispersion, ...)
 - Select a **distribution function** that is most suitable for explaining the behavior of the data.
 - Use the distribution function to **make predictions** about various properties of the population upon which the data are obtained.
- ❑ Measures of central tendency:
 - **Mean**: the average value of the set of data
 - **Median**: the value at the center of the set of data
 - **Mode**: the value that occurs most often
- ❑ Measures of dispersion:
 - Deviation: $d_i = x_i - \bar{x}$; mean deviation: $\bar{d} = \sum_{i=1}^n |d_i|/n$
 - **Standard deviation**: $\sigma = \sqrt{\sum_{i=1}^N \frac{(x_i - \mu)^2}{N}}$ (for population) or $S = \sqrt{\sum_{i=1}^n \frac{(x_i - \bar{x})^2}{n-1}}$ (for sample)
 - **Variance**: σ^2 (population) or S^2 (sample)

Example 1

- Find the mean, median, mode, range, standard deviation and variance of the following dataset

Solution

Compute **sample** measures with $n = 60$.

Mean $\bar{x} = \frac{x_1 + \dots + x_n}{n} = 1103^\circ\text{C}$

Median $x_m = 1104^\circ\text{C}$

Mode: $m = 1104^\circ\text{C}$

Range: $r = 1115 - 1089 = 26^\circ\text{C}$

Std. deviation $S = \sqrt{\sum_{i=1}^n \frac{(x_i - \bar{x})^2}{n-1}} = 5.79^\circ\text{C}^2$

Variance: $S^2 = 33.49^\circ\text{C}$

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Example 2: sample variance – divide by n or by $n - 1$?

- Suppose the population has only 3 members {1,2,3}

- The true mean is $\frac{1+2+3}{3} = 2$

- The true variance is $\frac{(1-2)^2 + (2-2)^2 + (3-2)^2}{3} = \mathbf{0.67}$

- Suppose we cannot view the whole population, but instead take a sample of size two.

- The next table shows the calculations of sample variance for all possible samples in two ways: dividing by $n - 1$ and dividing by n .

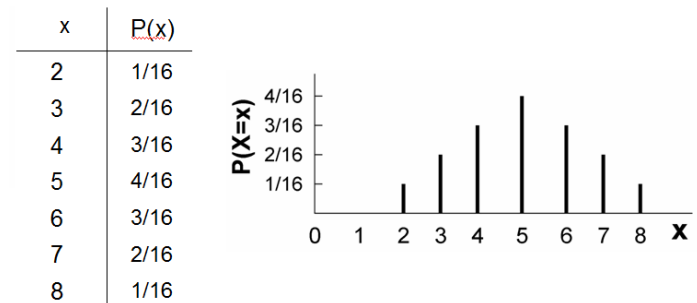
- Dividing by n results in an underestimate on the average across samples.

Sample	Sample mean	S^2 (div. n)	S^2 (div. $n-1$)
(1, 2)	1.5	0.25	0.5
(1, 3)	2	1	2
(2, 3)	2.5	0.25	0.5
(2, 1)	1.5	0.25	0.5
(3, 1)	2	1	2
(3, 2)	2.5	0.25	0.5
(1, 1)	1	0	0
(2, 2)	2	0	0
(3, 3)	3	0	0
Avg. across samples	2.5	0.33	0.67

Probabilistic functions (revisit)

□ Probability **mass** function (pmf)

- Normalization: $\sum_{i=1}^N P(x_i) = 1$
- Mean: $\mu = \sum_{i=1}^N x_i P(x_i)$
- Variance: $\sigma^2 = \sum_{i=1}^N (x_i - \mu)^2 P(x_i)$

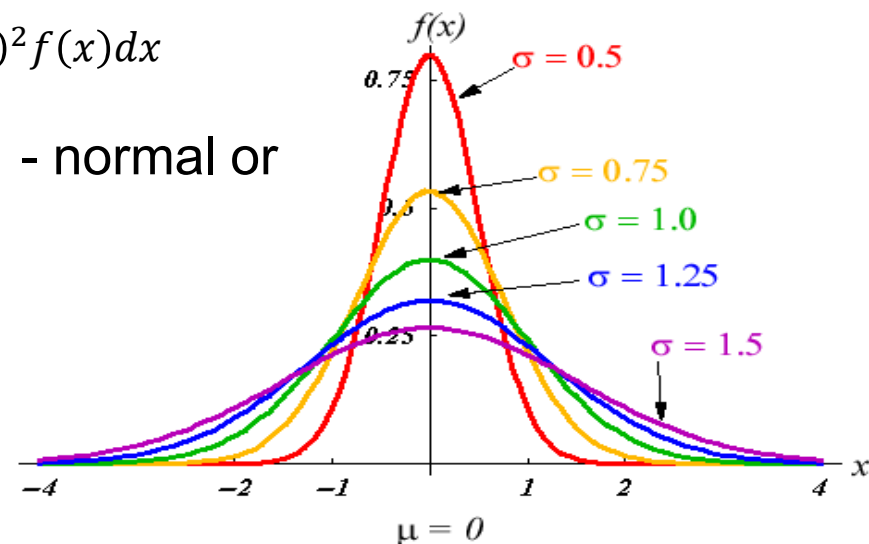


□ Probability **density** function (pdf)

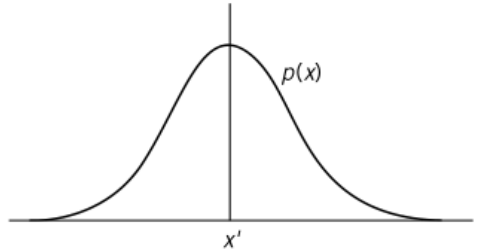
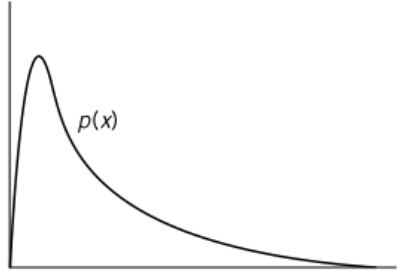
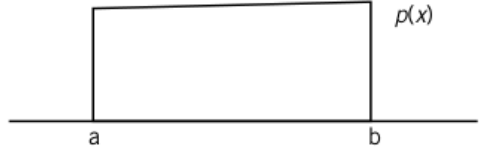
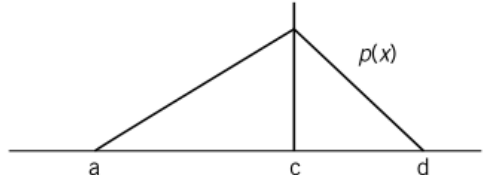
- $f(x_i)dx = P(x_i \leq x \leq x_i + dx)$; $P(a \leq x \leq b) = \int_a^b f(x)dx$
- Mean of population: $\mu = \int_{-\infty}^{+\infty} xf(x)dx$
- Variance of population: $\sigma^2 = \int_{-\infty}^{+\infty} (x - \mu)^2 f(x)dx$

□ **Example:** $f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-(x-\mu)^2/2\sigma^2}$ - normal or Gaussian distribution

- Bell-shaped, symmetric about μ
- σ defines the spread of the bell curve
- Confidence interval & confidence level
 $P(\mu - \sigma \leq x \leq \mu + \sigma) = 0.6826$



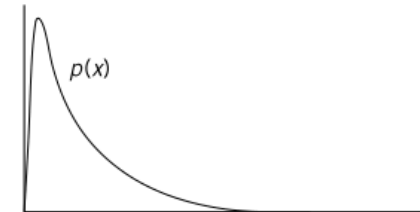
Frequently used distributions (1)

Distribution	Applications	Density Function	Shape
Normal	Most physical properties that are continuous or regular in time or space with variations due to random error	$p(x) = \frac{1}{\sigma(2\pi)^{1/2}} \exp \left[-\frac{1}{2} \frac{(x - x')^2}{\sigma^2} \right]$	
Log normal	Failure or durability projections; events whose outcomes tend to be skewed toward the extremity of the distribution	$p(x) = \frac{1}{\pi\sigma(2\pi)^{1/2}} \exp \left[-\frac{1}{2} \ln \frac{(x - x')^2}{\sigma^2} \right]$	
Rectangular	Processes in which a likely outcome falls in the range between minimum value a and maximum value b occurring with equal probability	$p(x) = \frac{1}{b - a} \text{ where } a \leq x \leq b, \text{ otherwise } p(x) = 0$	
Triangular	Process in which the likely outcome x falls between known lower bound a and upper bound b , with a peak value or mode c ; used when population information is sparse	$p(x) = \frac{2(x - a)}{(b - a)(c - a)} \text{ for } a \leq x \leq c$ $= \frac{2(b - x)}{(b - a)(b - c)} \text{ for } c < x \leq b$	

Frequently used distributions (2)

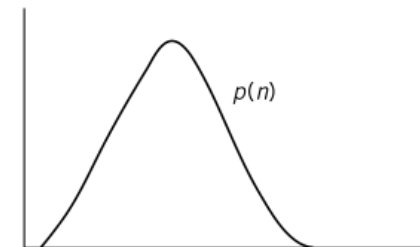
Poisson Events randomly occurring in time; $p(x)$ refers to probability of observing x events in time t ; here λ refers to x'

$$p(x) = \frac{e^{-\lambda} \lambda^x}{x!}$$



Binomial Situations describing the number of occurrences, n , of a particular outcome during N independent tests where the probability of any outcome, P , is the same

$$p(n) = \left[\frac{N!}{(N-n)!n!} \right] P^n (1-P)^{N-n}$$



Parameter estimation: from samples to population

- ❑ In most cases, sample size is small to the population, so how can parameters describing the **entire** population be estimated?
 - A (naïve) solution: estimating **population mean** by choosing many samples from population, find the mean of each sample and determine the uncertainty range of the means (**sample mean is also a random variable**) → very **costly**!
 - Desirable solution: estimate the population mean from one single sample mean & determine the confidence interval and confidence level.
- ❑ **Central limit theorem**: Take several different samples, each of size n , from the population of a RV x with mean μ & std. dev. σ
 - If n is **sufficiently large**, the sample means $\bar{x}$ follow a **normal distribution** with standard deviation $\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}}$, **regardless** of the distribution of the original population.
 - In most practical cases, to be considered large, n should exceed 30.
 - If the original population is **NOT** normal **AND** $n < 30$, the sample means $\bar{x}$ follow a normal distribution only approximately (will be discussed later).

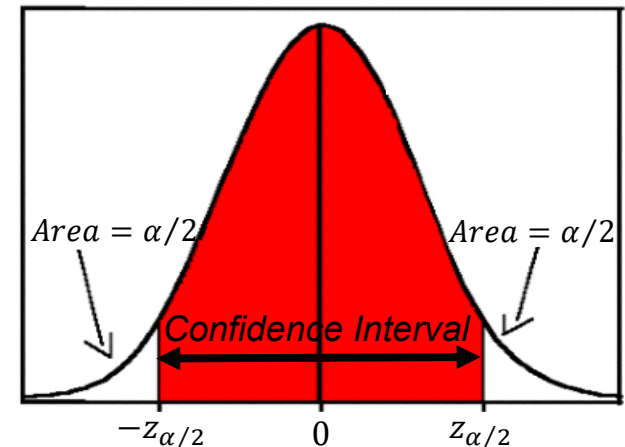
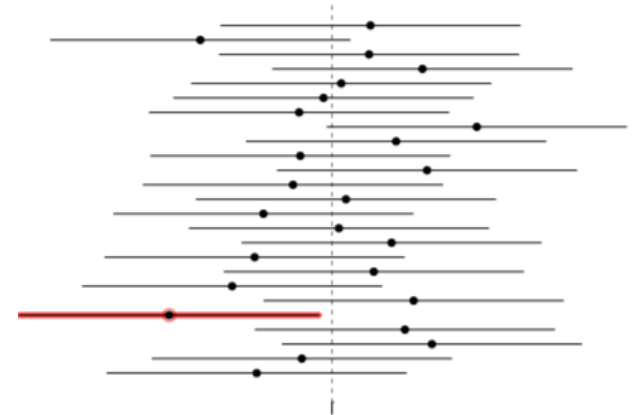
Interval estimation of population mean

- ❑ The CLT enables an estimate of the confidence interval for the population mean from a **large sample** with a suitable confidence level:

- Construct a RV from sample mean $z = \frac{\bar{x} - \mu}{\sigma_{\bar{x}}}$, according to CLT z is normally distributed $\rightarrow \mu$ has a confidence interval of $\bar{x}_i \pm z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$ with confidence level $1 - \alpha$.
- $\sigma \approx S$ (this approx. is **not** valid for small n).

- ❑ At the conf. level of $1 - \alpha$, we have

- $P\left(\bar{x}_i - z_{\alpha/2} \frac{\sigma}{\sqrt{n}} \leq \mu \leq \bar{x}_i + z_{\alpha/2} \frac{\sigma}{\sqrt{n}}\right) = 1 - \alpha$
- $P\left(\mu \leq \bar{x}_i + z_{\alpha} \frac{\sigma}{\sqrt{n}}\right) = 1 - \alpha$
- $P\left(\mu \geq \bar{x} - z_{\alpha} \frac{\sigma}{\sqrt{n}}\right) = 1 - \alpha$



Example

- ❑ Determine the 90% confidence interval of the mean resistance of a production batch from only one single sample of 36 readings where the average resistance is 25Ω and standard deviation is 0.5Ω .
- ❑ **Solution:**
 - Since $n = 36$ is considered sufficiently large $\rightarrow$ sample means follow normal distribution and the sample standard deviation, $S = 0.5\Omega$, can be used to approximate the batch standard deviation, σ , which we don't know.
 - $1 - \alpha = 90\% \rightarrow \alpha = 0.1$
 - $0.5 - \alpha/2 = 0.5 - 0.05 = 0.45$. Look up to the normal distribution table $\rightarrow z_{\alpha/2} = 1.645$.
 - $\bar{x} - z_{\alpha/2} \frac{S}{\sqrt{n}} = 25 - 1.645 \frac{0.5}{\sqrt{36}} = 24.86\Omega$; $\bar{x} + z_{\alpha/2} \frac{S}{\sqrt{n}} = 25 + 1.645 \frac{0.5}{\sqrt{36}} = 25.14\Omega$;
hence, $24.86\Omega \leq \mu \leq 25.14\Omega$ with 90% confidence .

Student's t-distribution

- ❑ If the sample size is small ($n < 30$), the distribution for the sample means $\bar{x}$ is only approximately normal

➤ $t = \frac{\bar{x} - \mu}{S/\sqrt{n}}$ follows Student's t-distribution

➤ Student's t-distribution function $f(t, \nu) = \frac{\Gamma(\frac{\nu+1}{2})}{\sqrt{\nu\pi}\Gamma(\frac{\nu}{2})\left(1+\frac{t^2}{\nu}\right)^{(\nu+1)/2}}$ depends on the number of samples because one degree of freedom is used (why?) to estimate the mean value of x , therefore $\nu = n - 1$.

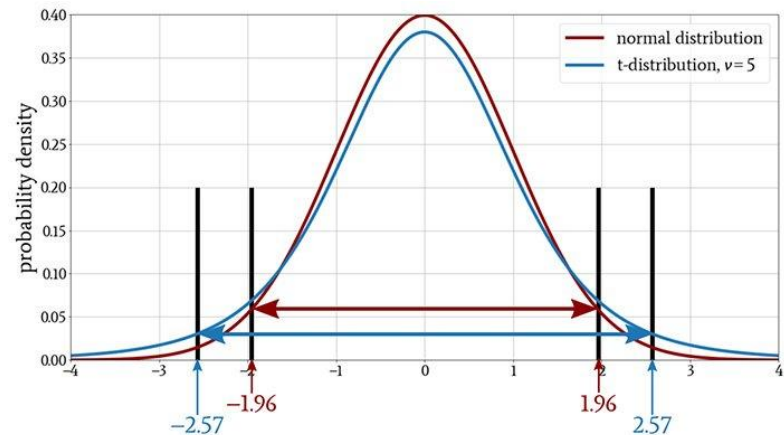
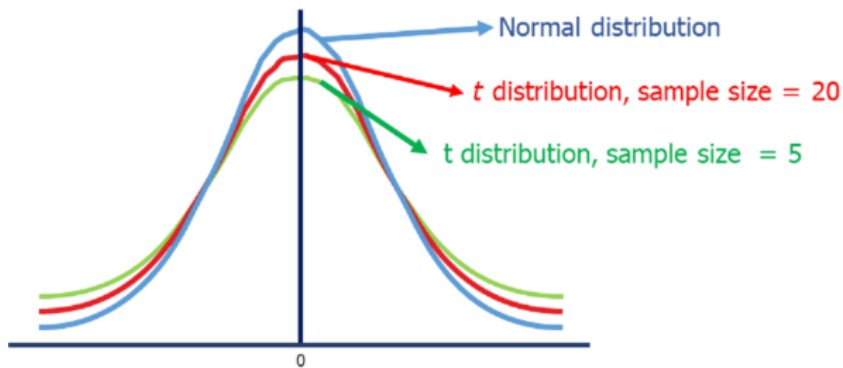


Table of Student's t-distribution

- Population mean confidence interval with confidence level α is $\bar{x}_i \pm t_{\alpha/2} \frac{s}{\sqrt{n}}$, i.e.

$$P\left(\bar{x}_i - t_{\alpha/2} \frac{s}{\sqrt{n}} \leq \mu \leq \bar{x}_i + t_{\alpha/2} \frac{s}{\sqrt{n}}\right) = 1 - \alpha.$$
- It is unlikely that an engineer would actually calculate $t_{\alpha/2}$ via the given formula
 → spreadsheet/statistical analysis programs are used instead.

v	$\alpha/2$				
	0.100	0.050	0.025	0.010	0.005
1	3.078	6.314	12.706	31.823	63.658
2	1.886	2.920	4.303	6.964	9.925
3	1.638	2.353	3.182	4.541	5.841
4	1.533	2.132	2.776	3.747	4.604
5	1.476	2.015	2.571	3.365	4.032
6	1.440	1.943	2.447	3.143	3.707
7	1.415	1.895	2.365	2.998	3.499
8	1.397	1.860	2.306	2.896	3.355
9	1.383	1.833	2.262	2.821	3.250
10	1.372	1.812	2.228	2.764	3.169
11	1.363	1.796	2.201	2.718	3.106
12	1.356	1.782	2.179	2.681	3.054
13	1.350	1.771	2.160	2.650	3.012
14	1.345	1.761	2.145	2.624	2.977
15	1.341	1.753	2.131	2.602	2.947

Example

- ❑ A manufacturer of mobile phones would like to estimate the mean failure time of a mobile phone brand with 95% confidence. Six handsets are tested with time to failure are: 1250, 1320, 1542, 1464, 1275, and 1383 hours. Estimate the population mean and the 95% CI.

- ❑ **Solution:**

- Because the sample size is small ($n = 6 < 30$), we should use the t-distribution to estimate the confidence interval.
- Sample mean and standard deviation: $\bar{x} = 1372$ h, $S = 114$ h.
- $1 - \alpha = 0.95 \Rightarrow \alpha = 0.05$. Look at the t-distribution table for $\nu = 6 - 1 = 5$, $\frac{\alpha}{2} = 0.025$, we obtain $t_{\alpha/2} = 2.571$. Hence, $\mu = \bar{x} \pm t_{\alpha/2} \frac{S}{\sqrt{n}} = 1372 \pm 120$ h.
- ❖ To reduce the 95% CI to ± 50 h, how many more handsets should be tested?
 - Suppose $n > 30$: $\bar{x} \pm z_{\alpha/2} \frac{S}{\sqrt{n}} = \bar{x} \pm 50 \Rightarrow n = \left(1.96 \times \frac{114}{50}\right)^2 = 20 \Rightarrow$ we have to use the t-distribution instead of the standard normal distribution.
 - Use $n = 20$ as an initial estimate, we obtain $t_{\alpha/2} = 2.093 \Rightarrow n = \left(2.093 \times \frac{114}{50}\right)^2 = 23$. Verify: take $n = 23$, recalculate $t_{\alpha/2}$ & $n \rightarrow n$ is unchanged.

Interval Estimation of the Population Variance

- ❑ The best estimate of the population variance, σ^2 , is the sample variance, S^2 .
 - $S^2 = \frac{\sum_{i=1}^n (x_i - \mu)^2}{n-1}$ is also a random variable (different from sample to sample).
 - Similar to the population mean, it is also necessary to establish a confidence interval for the estimated variance.

- ❑ For **normally distributed** populations, an RV $\chi^2 = \frac{\sum_{i=1}^n (x_i - \mu)^2}{\sigma^2}$ is used for the purpose of establishing a confidence interval.

- $\chi^2 = (n-1) \frac{S^2}{\sigma^2}$ (why?)
- $f(\chi^2) = \frac{(\chi^2)^{(v-2)/2} e^{-\chi^2/2}}{2^{v/2} \Gamma(v/2)}$, where one degree of freedom is used: $v = n - 1$.
- $P(\chi_{v,1-\alpha/2}^2 \leq \chi^2 \leq \chi_{v,\alpha/2}^2) = 1 - \alpha \rightarrow$ for a confidence level of $1 - \alpha$ the confidence interval on σ^2 is:

$$\frac{(n-1)S^2}{\chi_{v,\alpha/2}^2} \leq \sigma^2 \leq \frac{(n-1)S^2}{\chi_{v,1-\alpha/2}^2}$$

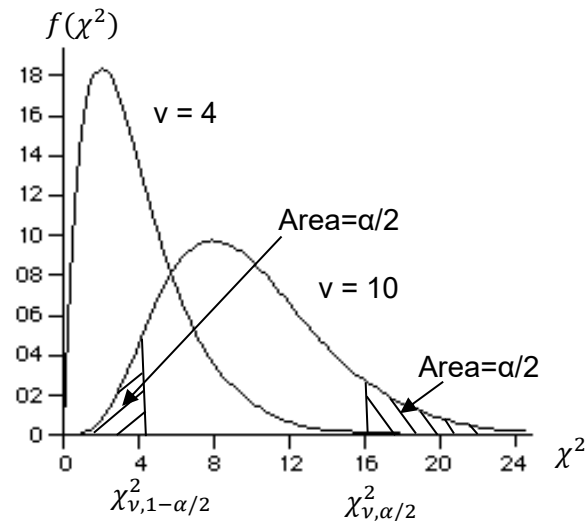


Table of Chi-squared distribution

$v \downarrow$	χ^2									
	Area of right hand tail					1 - $\alpha/2$ and $\alpha/2$				
	0.995	0.990	0.975	0.950	0.900	0.100	0.050	0.025	0.010	0.005
1	0.000039	0.000157	0.000982	0.003932	0.015791	2.706	3.841	5.024	6.635	7.879
2	0.0100	0.0201	0.0506	0.1026	0.2107	4.605	5.991	7.378	9.210	10.597
3	0.0717	0.1148	0.2158	0.3518	0.5844	6.251	7.815	9.348	11.345	12.838
4	0.2070	0.2971	0.4844	0.7107	1.0636	7.779	9.488	11.143	13.277	14.86
5	0.4118	0.5543	0.8312	1.1455	1.6103	9.236	11.070	12.832	15.086	16.750
6	0.6757	0.8721	1.2373	1.6354	2.2041	10.645	12.592	14.449	16.812	18.548
7	0.9893	1.2390	1.6899	2.1673	2.8331	12.017	14.067	16.013	18.475	20.278
8	1.3444	1.6465	2.1797	2.7326	3.4895	13.362	15.507	17.535	20.090	21.955
9	1.7349	2.0879	2.7004	3.3251	4.1682	14.684	16.919	19.023	21.666	23.589
10	2.1558	2.5582	3.2470	3.9403	4.8652	15.987	18.307	20.483	23.209	25.188
11	2.6032	3.0535	3.8157	4.5748	5.5778	17.275	19.675	21.920	24.725	26.757
12	3.0738	3.5706	4.4038	5.2260	6.3038	18.549	21.026	23.337	26.217	28.300
13	3.5650	4.1069	5.0087	5.8919	7.0415	19.812	22.362	24.736	27.688	29.819
14	4.0747	4.6604	5.6287	6.5706	7.7895	21.064	23.685	26.119	29.141	31.319
15	4.6009	5.2294	6.2621	7.2609	8.5468	22.307	24.996	27.488	30.578	32.801

- Find the interval boundary values of χ^2 for a 90% confidence level and $v = 10$?
- $\alpha = 1 - 0.9 = 0.1 \rightarrow$ the area of each tail is $\alpha/2 = 0.05 \rightarrow$ look up the upper value of χ^2 in the table under the column labeled 0.05 $\rightarrow \chi^2_{10,0.05} = 18.307$.
 - For the lower value of χ^2 , the area of the right-hand tail is $1 - \alpha/2 = 0.95 \rightarrow$ looking under the column labeled 0.95, we obtain $\chi^2_{10,0.95} = 3.9403$.

Example

- To estimate the uniformity of the diameter of ball bearings in a production batch, a sample of 20 is chosen and carefully measured. The sample mean is 0.32500 in, and the sample standard deviation is 0.00010 in. Obtain a 95% confidence interval for the standard deviation of the production batch.

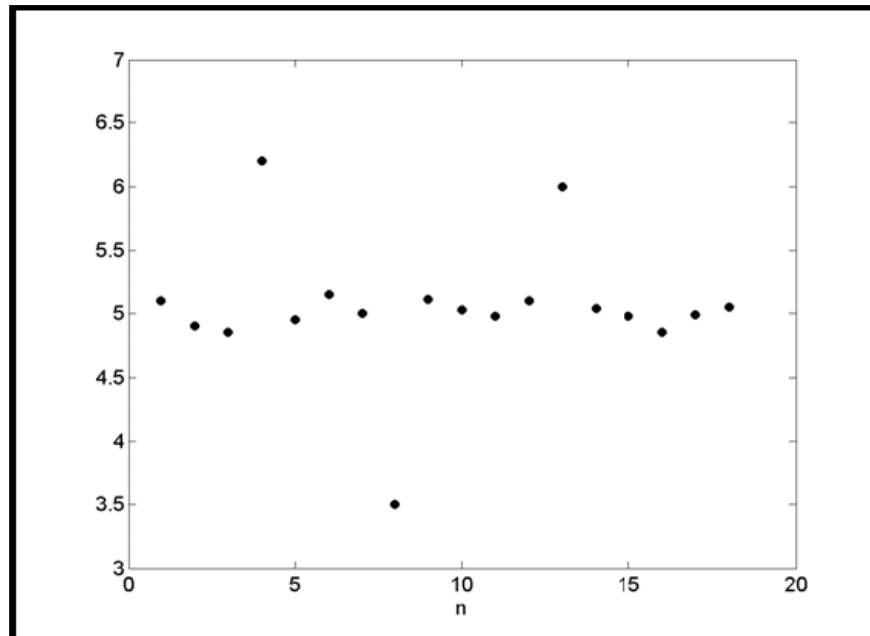
$v \downarrow$	χ^2 Area of right hand tail					$1 - \alpha/2$ and $\alpha/2$				
	0.995	0.990	0.975	0.950	0.900	0.100	0.050	0.025	0.010	0.005
16	5.1422	5.8122	6.9077	7.9616	9.3122	23.542	26.296	28.845	32.000	34.267
17	5.6973	6.4077	7.5642	8.6718	10.0852	24.769	27.587	30.191	33.409	35.718
18	6.2648	7.0149	8.2307	9.3904	10.8649	25.989	28.869	31.526	34.805	37.156
19	6.8439	7.6327	8.9065	10.1170	11.6509	27.204	30.144	32.852	36.191	38.582
20	7.4338	8.2604	9.5908	10.8508	12.4426	28.412	31.410	34.170	37.566	39.997

□ Solution:

- Assume that the population is normal, we have: $v = 19, \alpha/2 = 0.025$.
- Looking at the table for $v = 19, \alpha/2 = 0.025, 1 - \alpha/2 = 0.975$ yields $\chi^2_{19,0.025} = 32.852$ and $\chi^2_{19,0.975} = 8.907$.
- The interval for the batch standard deviation: $(20 - 1) \frac{0.0001^2}{32.852} \leq \sigma^2 \leq (20 - 1) \frac{0.0001^2}{8.907}$, or $0.000076 \leq \sigma \leq 0.00015$ in.

Criterion for Rejecting Questionable Data Points

- ❑ In some experiments, it happens that one or more measured data points appears to be out of the line with the rest of the data.
 - If some clear fault can be detected in measuring those specific values, they should be discarded, but often the seemingly faulty data cannot be traced to any specific problem.
 - Data that lie outside the probability of normal variation can bias the statistical analysis.



Statistical methods for rejecting questionable data points

- ❑ There are several statistical methods for detecting and rejecting these wild or **outlier** data points.
- ❑ Simplest method for the outlier detection is **three-sigma test**
 - First calculate the mean and the standard deviation from the data set, then label all the data points that lie outside the range of 99.7% probability of occurrence: $\bar{x} - 3\sigma \leq x \leq \bar{x} + 3\sigma$.
- ❑ Another method is **modified Thompson π technique**.
 - In this method, if we have n measurements that have a mean $\bar{x}$ and standard deviation S , the data can be arranged in ascending order. The extreme values (i.e. highest and lowest) are suspected outliers.
 - For these suspected points, the deviation is calculated as: $\delta_i = |x_i - \bar{x}|$.
 - In the next step, for the given value of n , the value of τ is obtained from Table 6.8. Compare the largest value of δ_i with the product of τ and standard deviation S : if $\delta_i > \tau S$, the data value x_i can be rejected as the outlier.
 - This method rejects one data value at a time, and is repeated with the recomputed values of mean and standard deviation of the remaining data until no more of the data can be eliminated.

The modified Thompson π technique

- ❑ It should be noted that eliminating an outlier is not an entirely positive event.
- ❑ The outlier may be resulted from a problem with the measuring system, the influence of extraneous variable or the measurement apparatus.

TABLE 6.8 Values of Thompson's τ

Sample size		Sample size	
n	τ	n	τ
3	1.150	22	1.893
4	1.393	23	1.896
5	1.572	24	1.899
6	1.656	25	1.902
7	1.711	26	1.904
8	1.749	27	1.906
9	1.777	28	1.908
10	1.798	29	1.910
11	1.815	30	1.911
12	1.829	31	1.913
13	1.840	32	1.914
14	1.849	33	1.916
15	1.858	34	1.917
16	1.865	35	1.919
17	1.871	36	1.920
18	1.876	37	1.921
19	1.881	38	1.922
20	1.885	39	1.923
21	1.889	40	1.924

Source: ASME (1998).

Example

- ❑ Nine voltage measurements: 12.02, 12.05, 11.96, 11.99, 12.10, 12.03, 12.00, 11.95 and 12.16V.

- ❑ **Solution:**

- ❑ First, calculate V_{mean} and S with 9 measurements:

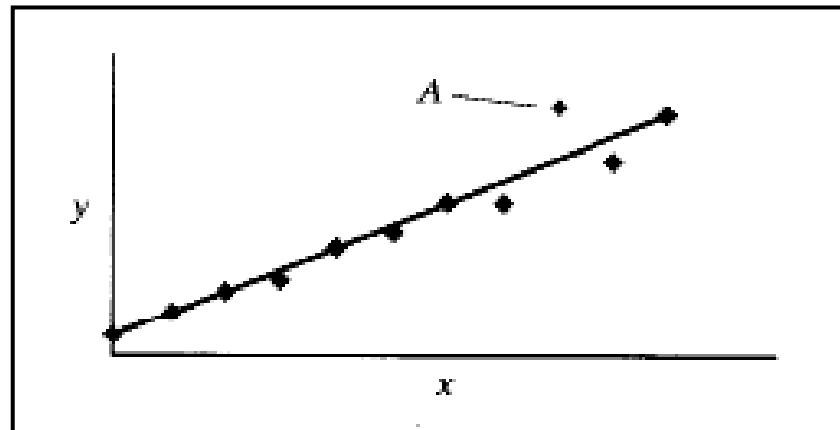
- $V_{\text{mean}} = 12.03\text{V}$, $S = 0.07$.
 - $D_1 = |V_{\text{largest}} - V_{\text{mean}}| = |12.16 - 12.03| = 0.13$.
 - $D_2 = |V_{\text{smallest}} - V_{\text{mean}}| = |11.95 - 12.03| = 0.08$.
 - From Table 6.8, $n = 9$, $\tau = 1.777 \rightarrow S\tau = 0.07 \times 1.777 = 0.12$.
 - $D_1 > S\tau$, 12.16 rejected.

- ❑ Now, re-calculate V_{mean} and S with 8 measurements.

- $V_{\text{mean}} = 12.01\text{V}$, $S = 0.05$.
 - $D_1 = |V_{\text{largest}} - V_{\text{mean}}| = |12.10 - 12.01| = 0.09$.
 - $D_2 = |V_{\text{smallest}} - V_{\text{mean}}| = |11.95 - 12.01| = 0.06$.
 - From Table 6.8, $n = 8$, $\tau = 1.749 \rightarrow S\tau = 0.05 \times 1.749 = 0.09$.
 - No rejection.

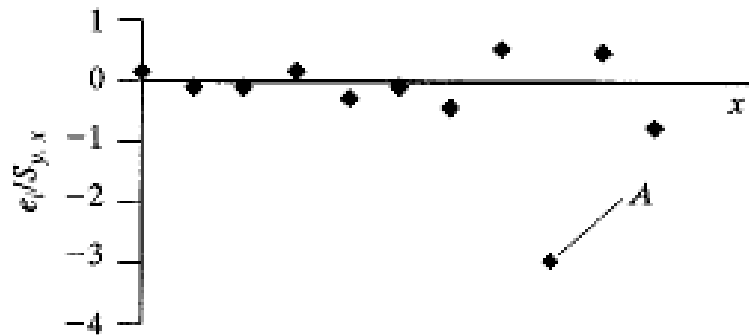
Outliers in x-y Data Sets

- ❑ We now know to detect outliers when static measurements are made, i.e. several measurements of a single variable.
 - How do we detect outliers when a variable y is measured as a function of an independent variable x ?
 - In most cases, there is only one value of y for each value of x .
- ❑ One way of identifying the outlier is to plot the data and the **best fit line** and identify the outlier that has much larger deviation from the line than the other data.



Outliers in x-y Data Sets

- ❑ A more sophisticated method of identifying outliers in x-y data set is by computing the ratio of the residuals e_i to the standard error of estimate S_{yx} , which is called the **standardized residual**.
 - If the residuals are normally distributed, it is expected that 95% of the standardized residuals would be in the range ± 2 . That is, within two standard deviations from the best-fit line.
 - If the standardized residual is much greater than 2 then it could be considered as the outlier.



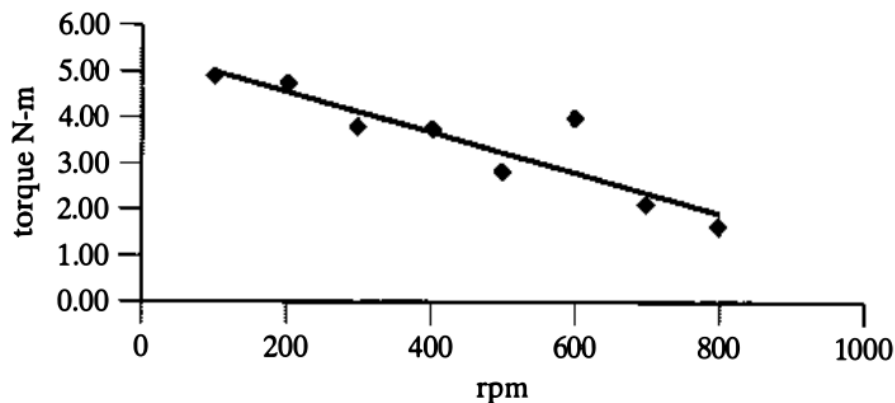
- ❑ Determination of outliers in x-y data sets is not a simple mechanistic process

Example - Outliers in x-y Data Sets

- ❑ Water-turbine experiment. Fit least squares straight line. Check outlier.

- ❑ **Solution:**

- Least squares straight line fit for the water turbine data: torque = 5.433 - 0.0044 rpm, $S_{yx} = 0.5758$.



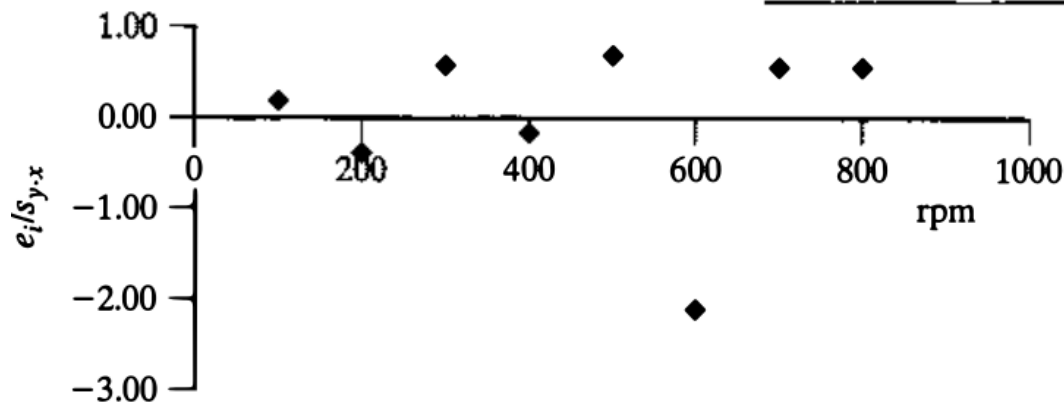
- The torque value at about 600 rpm appears to be inconsistent with the other data and is probably an outlier.

Speed (rpm)	Torque (Nm)
100	4.89
201	4.77
298	3.79
402	3.76
500	2.84
601	4
699	2.05
799	1.61

Example - Outliers in x-y Data Sets

- ❑ To confirm this, we can prepare a plot of the standardized residuals.

rpm	torque (measured)	torque (predicted)	$e/S_{y.x}$
100	4.89	4.99	0.18
201	4.77	4.55	-0.38
298	3.79	4.12	0.58
402	3.76	3.66	-0.16
500	2.84	3.23	0.68
601	4.00	2.79	-2.10
699	2.05	2.36	0.53
799	1.61	1.92	0.54



Summary

❑ Random error analysis

- General probabilistic concepts (revisit)
- Steps to perform statistical inferences

❑ Parameter estimation

- Interval estimation of population mean: Central limit theorem & Student's t distribution approximation
- Interval Estimation of the Population Variance: Chi-squared distribution

❑ Statistical methods for rejecting questionable data points

- Static measurements: three-sigma test, modified Thompson π technique.
- x-y Data Sets: standardized residual.

❑ **Next lecture:** (Total) uncertainty analysis.